

Extremal partition shapes for the order of $T(X, \mathcal{P})$

Which partition shape admits the most partition-preserving transformations

Abstract

Let X be a finite set with $|X| = n$ and let $\mathcal{P}$ be a partition of X into m blocks. Sarkar and Singh (2021) computed the order of the monoid $T(X, \mathcal{P})$ of partition-preserving full transformations for an arbitrary finite partition. That order depends only on the multiset of block sizes, so it is a function of the *shape* of $\mathcal{P}$ — a partition of n into m parts. We ask which shape maximises and which minimises it.

We prove that for fixed n and fixed $m \geq 2$, the order is **strictly increasing along the dominance order** on shapes. Consequently it is uniquely maximised at the most unequal shape $(n - m + 1, 1, \dots, 1)$ and uniquely minimised at the balanced shape. The same statement and the same proof hold for the natural r -ary generalisation at every arity $r \geq 1$, and for the submonoid of partial maps whose domain is a union of blocks.

The proof is short. The order is a product of factors $\sum_j n_j^k$, and the shape enters twice: once as the **bases** n_j , and once as the **exponents** $k = n_i$ at which those factors are evaluated. Separating the two roles, the base dependence is handled by Karamata's inequality and the exponent dependence by log-convexity of sums of exponentials. Both steps move the same way under a dominance-increasing exchange.

We also record a **negative result that delimits the phenomenon**. The analogous count for partial transformations *without* a condition on the domain — the object of Cicalò, Fernandes and Schneider (2012) — has factor function $\sum_j (n_j + 1)^k - (m - 1)$, a sum of exponentials *minus* a positive constant. Subtracting a constant does not preserve log-convexity, the proof fails, and the extremal law is **false**: at $n = 11$, $m = 9$ the maximiser and minimiser exchange places.

We claim no priority. Section 7 records what was and was not found in the literature, and which unobtained papers could overturn that.

1. The object

Let X be a finite set, $|X| = n$, and let $\mathcal{P} = \{X_1, \dots, X_m\}$ be a partition of X into nonempty blocks, $|X_i| = n_i$. Write $p = (n_1, \dots, n_m)$, a partition of n into exactly m positive parts; we call p the **shape** of $\mathcal{P}$ and always list it in weakly decreasing order.

Following Pei, a map $f \in \mathcal{T}_X$ **preserves** $\mathcal{P}$ if for every block X_i there is a block X_j with $X_i f \subseteq X_j$; equivalently $x \underset{\mathcal{P}}{\sim} y \implies x f \underset{\mathcal{P}}{\sim} y f$. The two formulations coincide for total maps, a point stated explicitly by Araújo and Schneider. Write

$$T(X, \mathcal{P}) = \{ f \in \mathcal{T}_X : (\forall X_i)(\exists X_j) X_i f \subseteq X_j \},$$

a submonoid of $\mathcal{T}_X$.

Proposition 1.1 (Sarkar–Singh, 2021, Thm. 6.1).

$$|T(X, \mathcal{P})| = \prod_{i=1}^m \sum_{j=1}^m n_j^{n_i}.$$

Remark. Sarkar and Singh state this grouped by size classes, as $\prod_i (\sum_j m_j n_j^{n_i})^{m_i}$ where there are m_i blocks of size n_i . The two forms are equal; regrouping over individual blocks is more convenient

here. The formula is immediate: each block X_i independently chooses a target block X_j and then an arbitrary map $X_i \rightarrow X_j$, of which there are $n_j^{n_i}$.

Since the right-hand side depends only on p , we write $\tau(p)$ for it.

The r -ary version

For $r \geq 1$ call an operation $f : X^r \rightarrow X$ **partition-preserving** if for every ordered r -tuple of blocks $T = (X_{i_1}, \dots, X_{i_r})$ there is a block X_j with $f(X_{i_1} \times \dots \times X_{i_r}) \subseteq X_j$. The same count applies with the product block $\prod_{i \in T} n_i$ in place of n_i :

$$\tau_r(p) = \prod_T \sum_{j=1}^m n_j^{e_T}, \quad e_T = \prod_{i \in T} n_i,$$

the product running over all m^r ordered r -tuples. At $r = 1$ this is Proposition 1.1. At $r = 0$ there is a single input tuple and $\tau_0(p) = n$ independent of the shape, so $r \geq 1$ is required throughout.

2. The question, and an example

τ_r is a function on the set of partitions of n into exactly m parts. That set carries the **dominance** (majorisation) order: $q \succ p$ when $\sum_{i \leq t} q_i \geq \sum_{i \leq t} p_i$ for every t , with equal totals. Its unique maximum is $(n - m + 1, 1, \dots, 1)$ and its unique minimum is the balanced shape, with parts differing by at most one.

Question. How does τ_r behave along this order?

All five shapes of $n = 8$ into $m = 3$ blocks, listed in decreasing dominance order:

shape	$\tau(p) = T(X, \mathcal{P}) $
(6, 1, 1)	2,986,112
(5, 2, 1)	757,920
(4, 3, 1)	248,768
(4, 2, 2)	165,888
(3, 3, 2)	84,568

Monotone, and the spread is not marginal. The ratio between extremes grows quickly:

n	m	argmax	argmin	ratio
8	3	(6, 1, 1)	(3, 3, 2)	35.3
10	4	(7, 1, 1, 1)	(3, 3, 2, 2)	249
12	4	(9, 1, 1, 1)	(3, 3, 3, 3)	4.92×10^3
16	8	(9, 1 ⁷)	(2 ⁸)	9.46×10^4
20	10	(11, 1 ⁹)	(2 ¹⁰)	1.39×10^7

3. The exchange lemma

Everything follows from a single local move.

Definition. Let p be a shape containing parts a and b with $a \geq b \geq 2$. The **exchange** at (a, b) replaces them by $a + 1$ and $b - 1$, producing a shape q of the same n with the same number of parts. Requiring $b \geq 2$ keeps the part from vanishing.

These moves generate the dominance order: $q \succ p$ if and only if q is reachable from p by a sequence of such exchanges, and every covering relation is of this form.

Theorem 3.1 (Exchange lemma). *For every $r \geq 1$ and every exchange with $a \geq b \geq 2$,*

$$\tau_r(q) > \tau_r(p).$$

Corollary 3.2 (Extremal shapes). *For fixed n and $m \geq 2$, τ_r is strictly increasing along the dominance order. It attains a unique maximum at $(n - m + 1, 1, \dots, 1)$ and a unique minimum at the balanced shape.*

Proof of the corollary from the theorem. A shape that is not balanced has two parts differing by at least 2, so it admits a reverse exchange strictly decreasing τ_r ; iterating terminates at the balanced shape. A shape other than $(n - m + 1, 1, \dots, 1)$ has a non-largest part of size at least 2, so it admits a forward exchange strictly increasing τ_r ; iterating terminates there. Uniqueness is uniqueness of the extremes of the dominance order. ◻

Note that "unique" means a unique **shape**, not a unique labelled partition of X .

4. Proof of the exchange lemma

4.1 The two roles of the shape

Write $S_k(p) = \sum_j n_j^k$ and

$$\tau_r(p) = \prod_T S_{e_T}(p).$$

The shape appears twice, and this is the whole difficulty: it supplies the **bases** n_j inside S_k , and it supplies the **exponents** e_T at which S_k is evaluated. An exchange changes both simultaneously. Attempts to compare $\tau_r(q)$ with $\tau_r(p)$ term by term fail because the two effects are entangled. The proof is to separate them and show that they move the same way.

4.2 Step 1: the bases

Lemma 4.1. *If $q \succ p$ then $S_k(q) \geq S_k(p)$ for every $k \geq 1$, with equality iff $k = 1$.*

Proof. $x \mapsto x^k$ is convex on $[0, \infty)$ and strictly convex for $k \geq 2$. Karamata's inequality applied to $q \succ p$ gives $\sum_j q_j^k \geq \sum_j p_j^k$, strictly for strictly convex x^k when the two multisets differ. At $k = 1$ both sides equal n . ◻

Hence $S_k(q) \geq S_k(p)$ **pointwise in k** , which is what lets the spectator factors be handled with no estimate at all.

4.3 Step 2: the exponents

Lemma 4.2. *For any fixed shape q , the function $k \mapsto S_k(q)$ is log-convex on $\mathbb{R}$.*

Proof. $S_k(q) = \sum_j e^{k \ln q_j}$ is a sum of exponentials with positive coefficients. For such a function, $(\ln S)''(k)$ is the variance of the values $\ln q_j$ under the probability weights q_j^k / S_k , hence nonnegative. ◻

This is the only analytic input in the argument, and it is worth being precise about what it does and does not require. A constant term is harmless: a constant c equals $c \cdot 1^k$, another exponential with positive coefficient and base 1. So $1 + S_k$ is log-convex exactly as S_k is, and neither the presence nor the absence of a constant term matters. What matters is that **every** term is a positive multiple of an exponential. Section 6 shows what goes wrong when one is not.

Lemma 4.3. *Let f be log-convex and positive, and let $a \geq b$. Then $f(a + 1)f(b - 1) \geq f(a)f(b)$.*

Proof. $h = \ln f$ is convex, so h' is nondecreasing in the sense that increments over intervals further right are larger. Since $a \geq b$, the interval $[a, a + 1]$ lies weakly to the right of $[b - 1, b]$, whence $h(a + 1) - h(a) \geq h(b) - h(b - 1)$. Rearranging and exponentiating gives the claim. ◻

4.4 Step 3: arity one

Let q be obtained from p by the exchange at (a, b) . Then

$$\tau(q) = \left(\prod_c S_c(q) \right) \cdot S_{a+1}(q) S_{b-1}(q),$$

where c runs over the parts of p other than the chosen occurrences of a and b — the **spectators**, which are the same for p and q . Now

$$\tau(q) \geq \left(\prod_c S_c(q) \right) S_a(q) S_b(q) > \left(\prod_c S_c(p) \right) S_a(p) S_b(p) = \tau(p),$$

the first inequality by Lemmas 4.2 and 4.3 applied to $f = S_\bullet(q)$, the second by Lemma 4.1 applied factor by factor. The second inequality is **strict**: the factor at exponent a satisfies $a \geq 2$, where Lemma 4.1 is strict. So no equality case can arise even if Step 2 is an equality. ■

4.5 Step 4: every arity, by tensoring

At arity r the exponents are the products e_T , and Step 2 must be replaced by

$$\sum_T \varphi(e_T(q)) \geq \sum_T \varphi(e_T(p)) \quad \text{for all convex } \varphi.$$

By Karamata, $(*)$ holds as soon as the exponent multiset $\{e_T(q)\}_T$ majorises $\{e_T(p)\}_T$. It does, and the reason is clean.

Lemma 4.4. *If $q \succ p$ then $\{e_T(q)\}_T \succcurlyeq \{e_T(p)\}_T$ for every $r \geq 1$.*

Proof. By Birkhoff–von Neumann, $q \succ p$ gives a doubly stochastic matrix D with $p = Dq$. Then $p^{\otimes r} = D^{\otimes r} q^{\otimes r}$, and a Kronecker product of doubly stochastic matrices is doubly stochastic. The coordinates of $p^{\otimes r}$, indexed by ordered r -tuples, are exactly the products e_T . A vector obtained from another by a doubly stochastic matrix is majorised by it. ◻

Applying $(*)$ with $\varphi = \ln S_\bullet(q)$, convex by Lemma 4.2, gives $\prod_T S_{e_T(q)}(q) \geq \prod_T S_{e_T(p)}(q)$, and Lemma 4.1 finishes as before. Strictness again comes from the all- a tuple, whose exponent $a^r \geq 2$. ■

Remark. A cellwise argument in the convex order also works — condition on which coordinates of the tuple land in the changed blocks, use closure of the convex order under multiplication by independent nonnegative factors, and sum over cells. It is longer. It is worth mentioning only because majorisation is **not** additive across such a decomposition while the inequality $(*)$ is, so the cellwise route must be phrased in the convex order; the tensor route above avoids the issue by never decomposing.

5. The partial-map submonoid with block-saturated domain

Let $PT_{\mathcal{P}}^{\text{sat}}(X)$ denote the partial maps t that preserve $\mathcal{P}$ in the sense of Section 1 wherever defined **and** whose domain is a union of blocks. This is a submonoid: if s sends block B into block B' , then B' is wholly inside or wholly outside $\text{dom } t$, so B survives or dies as a unit. Its idempotent partial identities are the id_D for D a union of blocks, forming a semilattice isomorphic to the Boolean lattice 2^m ; it is a left restriction submonoid of PT_X .

Proposition 5.1. $|PT_{\mathcal{P}}^{\text{sat}}(X)| = \prod_{i=1}^m \left(1 + \sum_{j=1}^m n_j^{n_i} \right).$

Proof. Each block independently is either wholly undefined, one way, or picks a target block and an arbitrary map into it. ◻

Proposition 5.2 (reduction to Proposition 1.1). *Adjoin a point $*$ and set $\mathcal{P}' = \mathcal{P} \cup \{\{*\}\}$ on $X \cup \{*\}$. The bijection "undefined $\mapsto *$ " identifies $PT_{\mathcal{P}}^{\text{sat}}(X)$ with the stabiliser of $*$ in $T(X \cup \{*\}, \mathcal{P}')$. Hence*

$$|PT_{\mathcal{P}}^{\text{sat}}(X)| = \frac{|T(X \cup \{*\}, \mathcal{P}')|}{n+1}.$$

Proof. Under the bijection a block either maps into a block of $\mathcal{P}$ or is sent wholesale to $*$, and $\{*\}$ is itself a block of $\mathcal{P}'$ — so block-saturation of the domain is exactly what makes the extended map partition-preserving. The block $\{*\}$ has $n + 1$ possible images, of which one fixes $*$. ◻

So Proposition 5.1 is a corollary of Proposition 1.1, not an independent count. Its factor function $1 + S_k$ is a sum of exponentials, so Theorem 3.1 and Corollary 3.2 apply verbatim, at every arity.

Terminology. The domain condition — if a point lies in the domain, so does its whole block — is not new. It appears in the literature on inverse semigroups as **$\mathcal{P}$ -stability**, for *injective, order-preserving* partial maps; see Section 7 for the attribution and its evidential status.

6. A negative result: what happens without the domain condition

Drop the condition on the domain and keep only agreement on successors where defined. This is the monoid PT_E of Cicalò, Fernandes and Schneider, whose order for a **uniform** partition into m blocks of size n they give as $(m(n+1)^n - m + 1)^m$. For an arbitrary shape the same reasoning gives

$$\beta(p) = \prod_{i=1}^m \left(1 + \sum_{j=1}^m [(n_j + 1)^{n_i} - 1] \right) = \prod_{i=1}^m \left(\sum_{j=1}^m (n_j + 1)^{n_i} - (m - 1) \right).$$

The factor function is now a sum of exponentials **minus a positive constant**. That is precisely the hypothesis of Lemma 4.2 failing: adding a positive constant is harmless because a constant is an exponential with base 1, but subtracting one is not, because $-c$ is not a positive multiple of an exponential.

Example 6.1. At $p = (1, 1)$ the factor function is $2^{k+1} - 1$, so $f(1), f(2), f(3) = 3, 7, 15$ and

$$f(1)f(3) = 45 < 49 = f(2)^2,$$

so f is not log-convex.

And the conclusion fails with it.

Example 6.2 (the extremal law is false for β). At $n = 11, m = 9$,

$$\beta(3, 1^8) = 120 \cdot 12^8 = 51,597,803,520, \quad \beta(2, 2, 1^7) = 38^2 \cdot 12^7 = 51,741,130,752.$$

The two shapes are comparable in dominance with $(3, 1^8) \succ (2, 2, 1^7)$, yet β is larger at the *less* unequal one. Since τ and $|PT^{\text{sat}}|$ are both maximised at $(3, 1^8)$ and minimised at $(2, 2, 1^7)$ in this cell, the maximiser and minimiser **exchange places** when the domain condition is dropped.

Controlled comparison. Holding the bases fixed at $n_j + 1$ and deleting only the subtracted constant isolates the cause. Over all shapes with $n \leq 26$ (42,903 distinct exchanges) and log-convexity probes over all shapes with $n \leq 12$ (1,482 instances):

factor function	log-convexity failures	exchange-lemma failures
$\sum_j (n_j + 1)^k$ — no subtraction	0 / 1,482	0 / 42,903
$\sum_j (n_j + 1)^k - (m - 1)$ — the monoid PT_E	292 / 1,482	66 / 42,903
$\sum_j n_j^k$ — Proposition 1.1	0 / 1,482	0 / 42,903
$1 + \sum_j n_j^k$ — Proposition 5.1	0 / 1,482	0 / 42,903

Rows 1 and 2 differ only by the subtraction and differ in outcome; rows 3 and 4 differ only by an added constant and do not.

The implication runs one way only. Failure of log-convexity at the exchanged exponents is *necessary* for an exchange to fail — across all 42,903 exchanges, every one of the 66 losses lies among the 3,871 log-convexity failures, and none outside. It is far from sufficient: only $66/3871 \approx 1.7\%$ of log-convexity failures produce an actual loss, since the spectator factors can absorb the deficit. A characterisation of the failing cells is **open**.

7. Position in the literature

7.1 What is known

Source	Object	Total/partial	Domain condition	Result
Pei, <i>Semigroup Forum</i> 49 (1994), 49–58	introduces $T_E(X)$	total	—	lattices of T -equivalences, α -congruences; no cardinality
Araújo–Schneider, arXiv:0807.1214	$T(X, \mathcal{P})$, uniform	total	—	rank
Araújo–Bentz–Mitchell–Schneider, arXiv:1404.1598	$T(X, \mathcal{P})$, arbitrary	total	—	rank, as a function of shape
Cicalò–Fernandes–Schneider, arXiv:1210.4775	PT_E , uniform	partial	none	order $(m(n+1)^n - m+1)^m$
Sarkar–Singh , <i>Comm. Algebra</i> 49 (2021), 331–342	$T(X, \mathcal{P})$, arbitrary	total	—	order ; our Proposition 1.1
Pei–Zhou, <i>Adv. Math. (China)</i> (2009)	$P_E(X)$, arbitrary E	partial	none	Green's relations, regularity; no cardinality
Fernandes, <i>Semigroup Forum</i> 56 (1998), 418–433	normally ordered inverse semigroups	partial, injective	union of blocks	origin of $\mathcal{P}$ -stability
Caneco–Fernandes–Quinteiro, arXiv:1905.11489	$POI_{k \times m}$	partial, injective	union of blocks	order, rank, presentation
Sun, <i>Bull. Malays. Math. Sci. Soc.</i> 36 (2013), 179–192	$OP_E(X)$, uniform	total	—	order and idempotent count

7.2 What was not found

No extremal or majorisation result for the order of any of these monoids as a function of partition shape. The literature studies **rank** as a function of shape — Araújo–Bentz–Mitchell–Schneider give the exact rank of $T(X, \mathcal{P})$ for arbitrary $\mathcal{P}$ — and computes **order** only for a partition that is fixed in advance.

There is a structural reason, and it is worth stating because it explains the absence without appealing to novelty. **Almost the entire literature on these monoids assumes a uniform partition.** Cicalò–Fernandes–Schneider, Araújo–Schneider, Caneco–Fernandes–Quinteiro, Fernandes–Quinteiro and Sun all fix m blocks of size n by hypothesis; under that hypothesis the shape is determined by (m, n) and the extremal question cannot be posed. The order for an arbitrary shape appears to have become available only with Sarkar–Singh in 2021. The question asked here is a natural one to ask of that formula, and we do not claim to be the first to ask it — only to record that a search did not find it asked.

Searches returning nothing relevant: "saturated domain", "block-saturated", " $\mathcal{P}$ -saturated" partial transformations; partition-preserving combined with non-uniform; extremal or Schur-convexity results for transformation-monoid orders. The extremal literature that surfaces on these terms — Hwang–Rothblum and related work on Schur-convex bounded-shape partition problems — concerns partitioning *numbers*, not counting partition-preserving maps. **OEIS**: no hits for the value sequences of τ or of Proposition 5.1, nor for Cicalò–Fernandes–Schneider's own published table; the only match is A014566 ($n^n + 1$), the degenerate one-block case.

7.3 Papers not obtained, and what would overturn the above

Honesty about sources is part of the claim. Three papers were sought and the outcomes differ:

Paper	Outcome	Bearing
Sun 2013 , BMMS 36(1), 179–192	Obtained in full (open access, EMIS). Counts $OP_E(X)$, $O_E(X)$ and their idempotents for a uniform partition, with Fibonacci numbers F_{2n} in the idempotent formulas. No shape dependence, no extremal statement	Does not overturn
Pei 1994 , <i>Semigroup Forum</i> 49, 49–58, DOI 10.1007/BF02573470	Not obtained — Springer paywalled, EuDML copy unreachable. zbMATH review obtained in full (Zbl 0804.20046), which quotes the definition $T_E(X) = \{f \in \mathcal{T}_X : (f(a), f(b)) \in E \forall (a, b) \in E\}$ and summarises the contents as lattices of T -equivalences and α -congruences	Low risk. The reviewed content contains no cardinality of any kind
Fernandes 1998 , <i>Semigroup Forum</i> 56, 418–433, DOI 10.1007/PL00005955	Not obtained. No preprint; the author's own publication page lists the citation without a file	The attribution of $\mathcal{P}$-stability to this paper is not verified from the primary source. It rests on the restatement in Caneco–Fernandes–Quinteiro, arXiv:1905.11489

An earlier draft of this note cited Fernandes 1998 as volume 58; the author's publication page and the DOI record both give **volume 56**, and that is used above.

What would overturn "no published extremal result". A counting result in the Chinese-language literature on these semigroups — *Advances in Mathematics (China)*, *J. Guizhou Normal Univ.*, *Xinyang Normal Univ. J.* — which is poorly indexed and was largely inaccessible; the Pei–

Zhou 2009 full text, of which only a database abstract was read; or any treatment of $T(X, \mathcal{P})$ for non-uniform $\mathcal{P}$ predating Sarkar–Singh.

8. Coverage boundary

The theorem is a statement about a specific counting measure, and it is false for several natural neighbours. These are recorded because they delimit it.

Class of operations counted	Extremal law
All maps $X^r \rightarrow X$, $r \geq 1$ (Prop. 1.1 and its r -ary form)	Theorem 3.1
Partial maps with block-saturated domain (Prop. 5.1)	Theorem 3.1
Nullary, $r = 0$	Vacuous: $\tau_0 = n$, shape-independent
Partial maps, no domain condition (PT_E)	False. Example 6.2
Bijections only	False for the minimum. At $n = 4$, $m = 2$, the shape $(2, 2)$ admits 8 partition-preserving permutations against 6 for $(3, 1)$
Maps of a prescribed domain size	False. At $n = 6$, $m = 3$, domain size 4, the shape $(2, 2, 2)$ admits 432 against 258 for $(4, 1, 1)$
Idempotents only	Open — see below

The idempotent case

For idempotent partial maps with block-saturated domain, write $S_i = \text{Fix}(t) \cap X_i$. A block carrying fixed points must target *itself*, since its fixed points map to themselves; a fixed-point-free block in the domain may target any block that carries some. Hence

$$\sigma^{\text{idem}}(p) = \sum_{D, (s_i)_{i \in D}} \prod_{i \in D} \binom{n_i}{s_i} \cdot \prod_{\substack{i \in D \\ s_i \geq 1}} s_i^{n_i - s_i} \cdot \prod_{\substack{i \in D \\ s_i = 0}} \left(\sum_{\substack{j \in D \\ s_j \geq 1}} s_j^{n_i} \right),$$

validated against exhaustive enumeration for all 29 shapes with $n \leq 6$.

The last bracket is the obstruction: a fixed-point-free block's factor depends on which *other* blocks carry fixed points. The count is a sum over configurations, not a product of per-block factors, so there is no factor function for log-convexity to be a property of and the method of Section 4 has nothing to act on. The extremal law nevertheless holds in every case computed (466 exchanges, $n \leq 13$). We state it as a conjecture and note that a proof would need a different mechanism.

9. Evidence grades

Stated separately, because the statements here are not all of the same kind.

Statement	Grade
Proposition 1.1	Published (Sarkar–Singh 2021). Independently checked here against exhaustive enumeration for all shapes with $n \leq 6$
Propositions 5.1, 5.2	Written proof; the identity of 5.2 verified exactly for all 507 shapes with $n \leq 14$
Theorem 3.1, Corollary 3.2	Written proof , not formalised. Every step machine-checked in exact integer arithmetic: arity 1 over 128,121 exchanges on 28,622 shapes; arities 2–5 for the majorisation step; direct verification of the conclusion at arities 1–4
Independent audit	An independent reviewer confirmed the proof on the stated domain and rejected three surrounding explanatory claims, since corrected; in particular an earlier draft wrongly asserted that the constant term is essential to Lemma 4.2
Examples 6.1, 6.2 and the table in §6	Exact finite computation, complete enumeration in the stated ranges
Coverage boundary rows	Exact counterexamples, retained
Idempotent formula	Written derivation + exhaustive validation, $n \leq 6$
Idempotent extremal law	Conjecture , finite test only
§7.2, "not found in the literature"	A recorded search outcome, not a novelty claim. §7.3 lists what was not read and what would overturn it

All computations are in exact integer or rational arithmetic; no floating point is used in any verification. Reproduction scripts accompany this note.

References

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